

DATA STRUCTURE

LAB PROGRAMS

18. Hashing

```
#include <stdio.h>

#define SIZE 10

int main() {

    int hashTable[SIZE], n, key, index;

    for(int i = 0; i < SIZE; i++)

        hashTable[i] = -1;

    printf("Enter number of elements: ");

    scanf("%d", &n);

    printf("Enter elements: ");

    for(int i = 0; i < n; i++) {

        scanf("%d", &key);

        index = key % SIZE;

        while(hashTable[index] != -1)

            index = (index + 1) % SIZE;

        hashTable[index] = key;

    }

    printf("Hash Table:\n");

    for(int i = 0; i < SIZE; i++)

        printf("%d --> %d\n", i, hashTable[i]);

    return 0;

}
```

Output

```
Enter number of elements: 5
Enter elements: 10 15 20 25 30
Hash Table:
0 --> 10
1 --> 20
2 --> 30
3 --> -1
4 --> -1
5 --> 15
6 --> 25
7 --> -1
8 --> -1
9 --> -1
```